Security Onion Project

INTRODUCTION:

Security Onion is a free and open-source Linux distribution focused on network security monitoring, intrusion detection, and log management. It is a complete solution for the detection, analysis, and response to various cyber threats on the network of an organization. Security Onion incorporates robust security tools such as Suricata for intrusion detection systems, Zeek for network traffic analysis, the Elastic Stack for log management, and Wazuh for host-based intrusion detection systems. It captures network traffic and analyzes it in real-time, enabling security experts to detect suspicious activities on the network and improve the overall security posture of the network. Security Onion is a powerful tool for Security Operation Centers (SOCs), enterprise networks, and cybersecurity training environments. The project focuses on the implementation of Security Onion in a virtual lab environment for simulating attacks and carrying out incident analysis.

Problem Statement:

In today's computerized world, organizations depend on computer networks and internet-based technologies for their day-to-day activities. However, the rapid development of computer technologies has also resulted in the emergence of various cyber threats and security incidents such as malware attacks, unauthorized access, brute-force attacks, phishing, and network intrusion. Traditional security solutions such as firewalls and antivirus solutions are not capable of detecting sophisticated and modern types of cyber threats and security incidents in real-time. Moreover, many organizations do not have proper network monitoring solutions to identify security incidents before they occur.

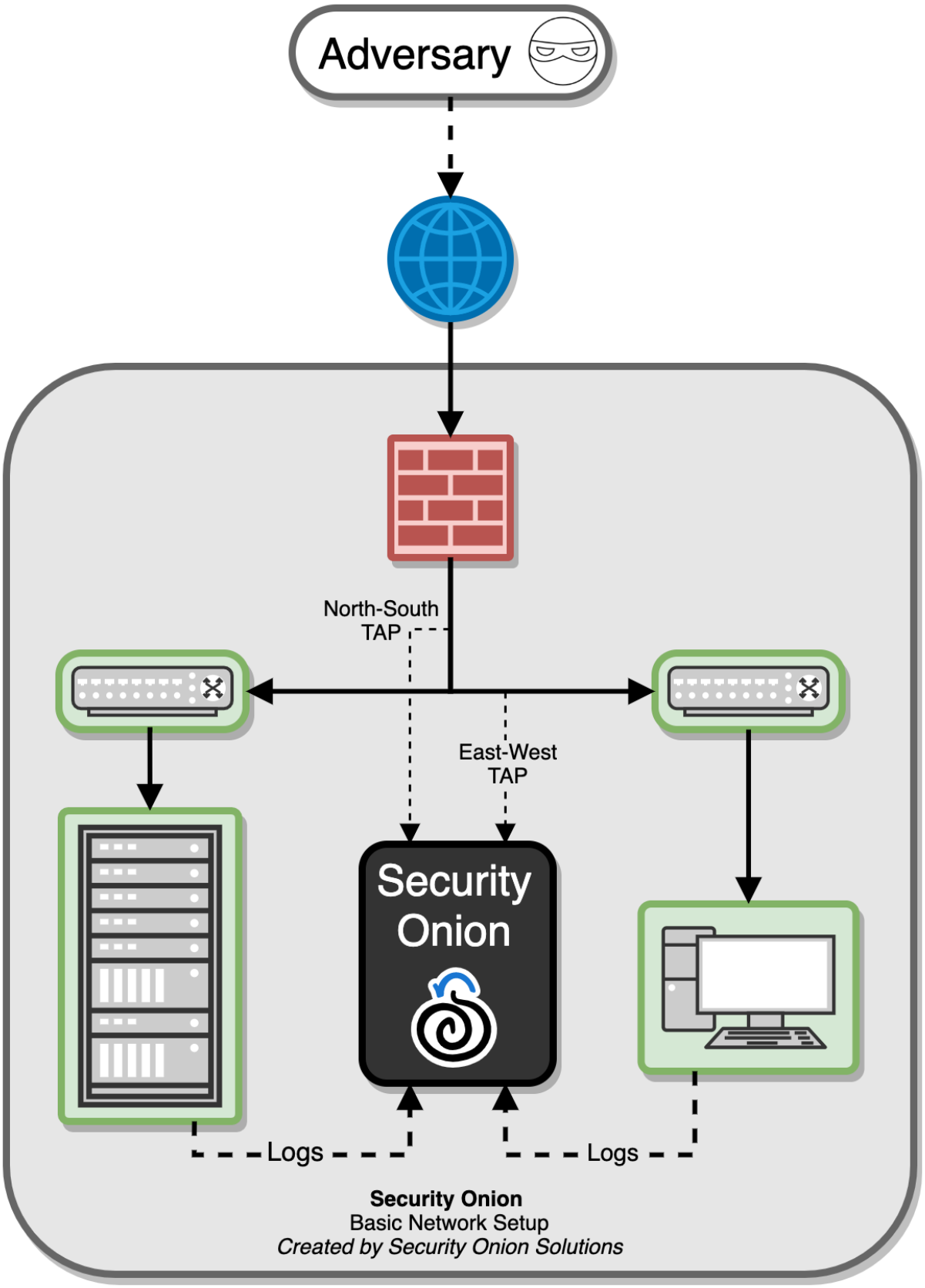
There is a need to design and implement an effective Network Security Monitoring and Intrusion Detection System to monitor network activities and identify security incidents. This project is designed to resolve the problem of network security monitoring and intrusion detection. This project will implement the Security Onion as an effective security solution to enhance network security monitoring and intrusion detection capabilities in a controlled environment.

Objectives of the Project:

The main objective of this project is to design and implement a Network Security Monitoring and Intrusion Detection system using Security Onion in a virtual lab environment. The specific objectives of the project are as follows:

- To deploy and configure Security Onion in a virtualized environment for network monitoring.
- To simulate real-world cyber attacks such as port scanning and brute-force attacks.
- To detect and analyze intrusions using Suricata and Zeek.
- To collect, store, and analyze logs using the Elastic Stack dashboard.
- To monitor alerts and investigate suspicious activities through the SOC interface.
- To generate incident reports and recommend mitigation strategies.
- To gain practical knowledge of Security Operations Center (SOC) workflow and threat detection techniques.

INSTALLATION:

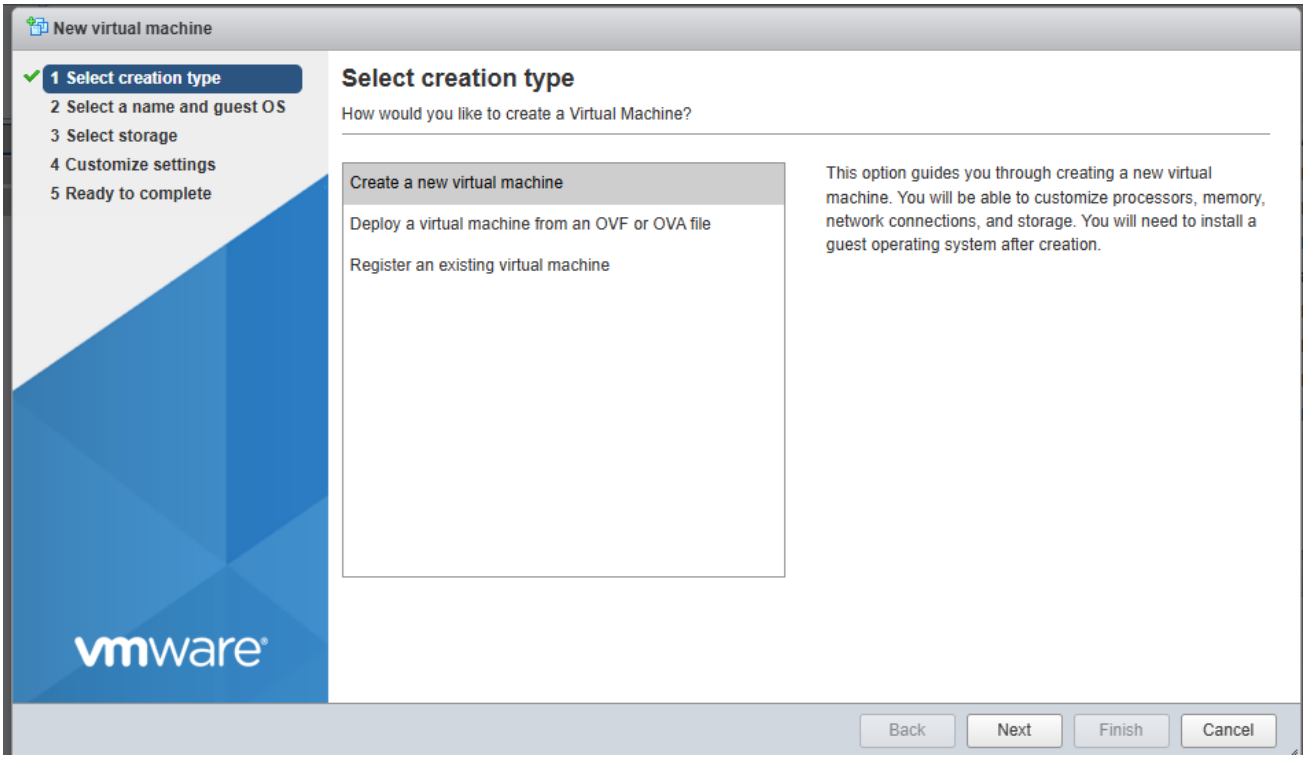


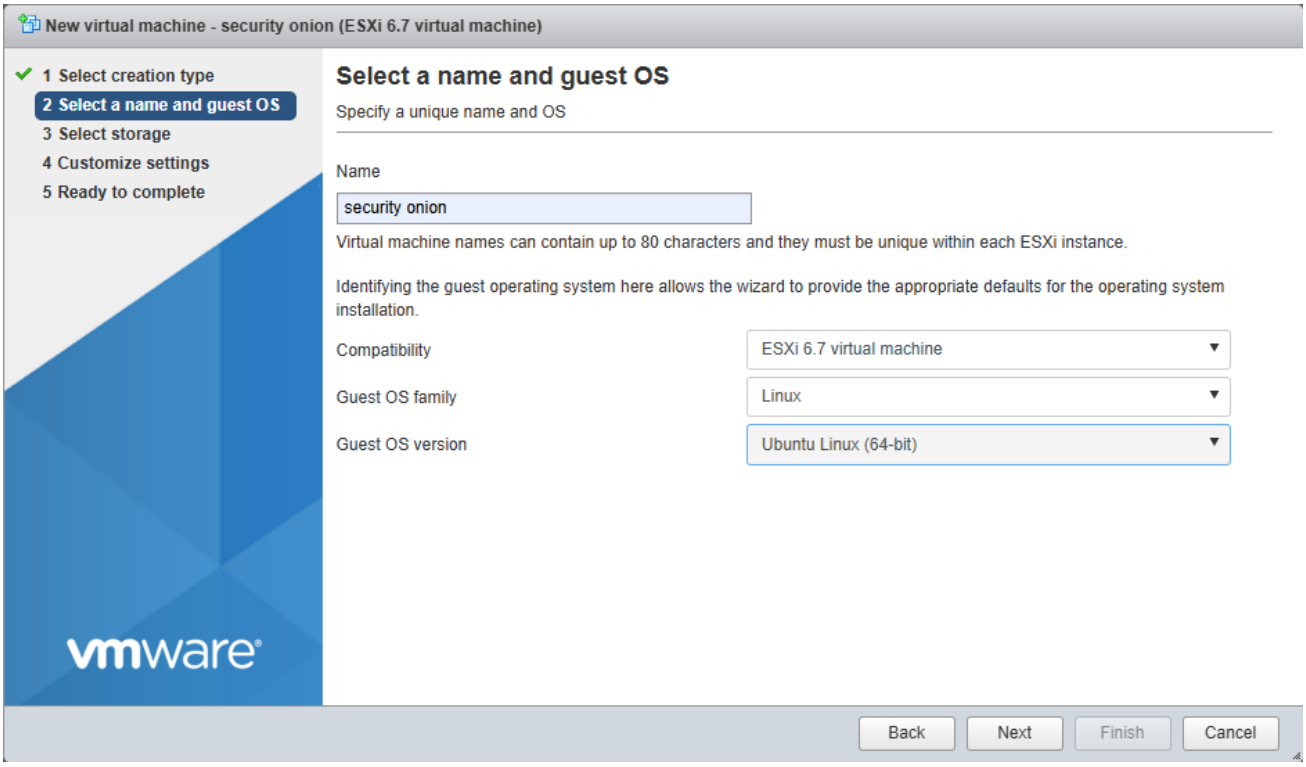
STEP 1:

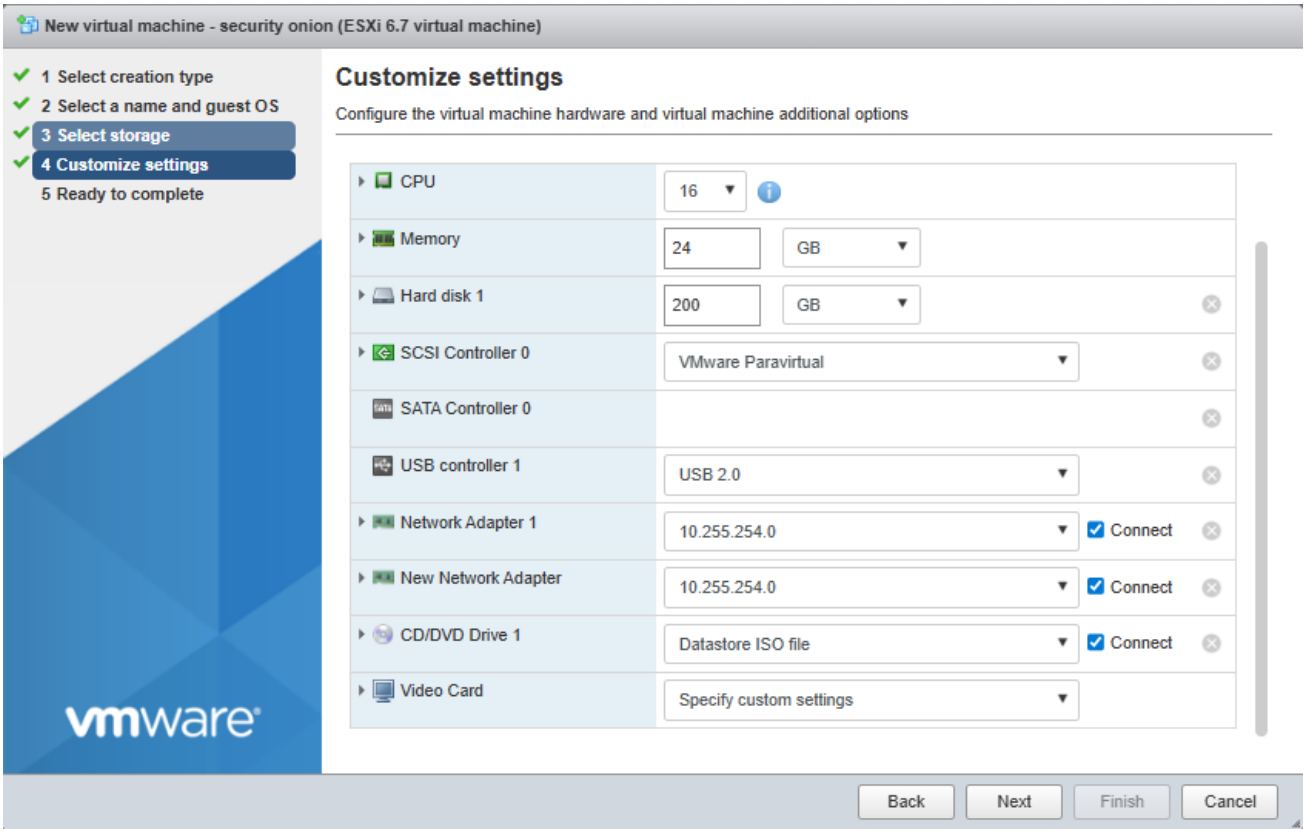
Download the Security Onion ISO File

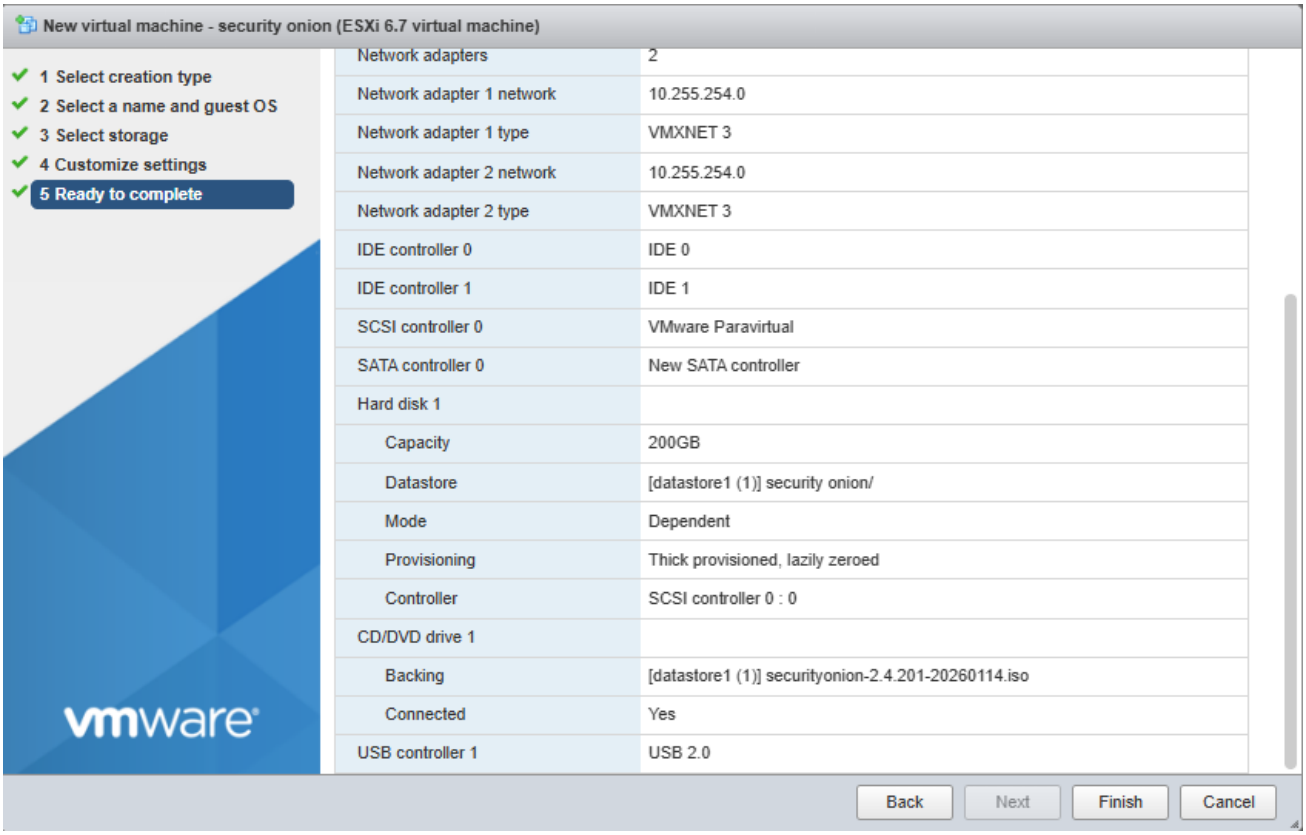
To begin the installation of Security Onion, you must first download the official ISO image (installation file) from the official source. Also in network adapter option add two network adapter

Link: https://github.com/Security-Onion-Solutions/securityonion/releases/tag/2.4.210-20260302






New virtual machine - security onion (ESXi 6.7 virtual machine)
1 Select creation type
2 Select a name and guest OS
3 Select storage
4 Customize settings
5 Ready to complete
Customize settings
Configure the virtual machine hardware and virtual machine additional options
CPU
16
Memory
24
GB
Hard disk 1
200
GB
SCSI Controller 0
VMware Paravirtual
SATA Controller 0
USB controller 1
USB 2.0
Network Adapter 1
10.255.254.0
Connect
New Network Adapter
10.255.254.0
Connect
CD/DVD Drive 1
Datastore ISO file
Connect
Video Card
Specify custom settings
vmware
Back
Next
Finish
Cancel


New virtual machine - security onion (ESXi 6.7 virtual machine)
1 Select creation type
2 Select a name and guest OS
3 Select storage
4 Customize settings
5 Ready to complete
Network adapters 2
Network adapter 1 network 10.255.254.0
Network adapter 1 type VMXNET 3
Network adapter 2 network 10.255.254.0
Network adapter 2 type VMXNET 3
IDE controller 0 IDE 0
IDE controller 1 IDE 1
SCSI controller 0 VMware Paravirtual
SATA controller 0 New SATA controller
Hard disk 1
Capacity 200GB
Datastore [datastore1 (1)] security onion/
Mode Dependent
Provisioning Thick provisioned, lazily zeroed
Controller SCSI controller 0 : 0
CD/DVD drive 1
Backing [datastore1 (1)] securityonion-2.4.201-20260114.iso
Connected Yes
USB controller 1 USB 2.0
vmware
Back
Next
Finish
Cancel

STEP 2:

Booting the ISO and Starting Installation

After downloading the Security Onion ISO file, the ISO was mounted in a virtual machine. Upon booting, the installation menu appeared with multiple options. The standard installation option “Install Security Onion 2.4.201” was selected to proceed with the setup. This option installs the complete Security Onion platform required for network monitoring and intrusion detection in a standalone lab environment.

STEP 3:

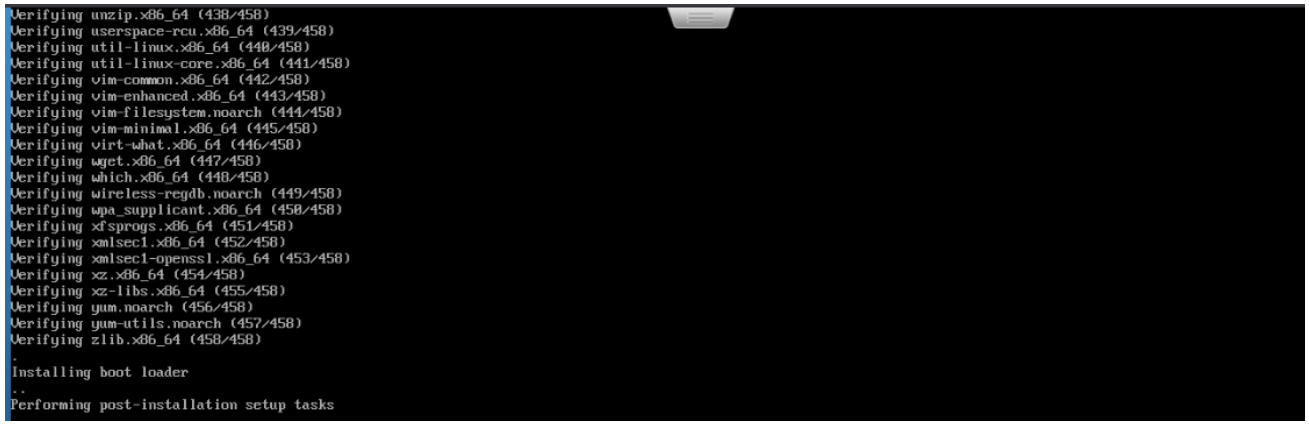
Creating Administrative User

During installation, the system prompted for confirmation before erasing all disk data. After typing “yes,” a new administrative user account was created for managing Security Onion. The username “socadmin” was configured, and a strong password was set to ensure secure access to the system. This account will be used for logging into the Security Onion management interface and performing administrative tasks.

STEP 4:

Installation Completion and System Reboot

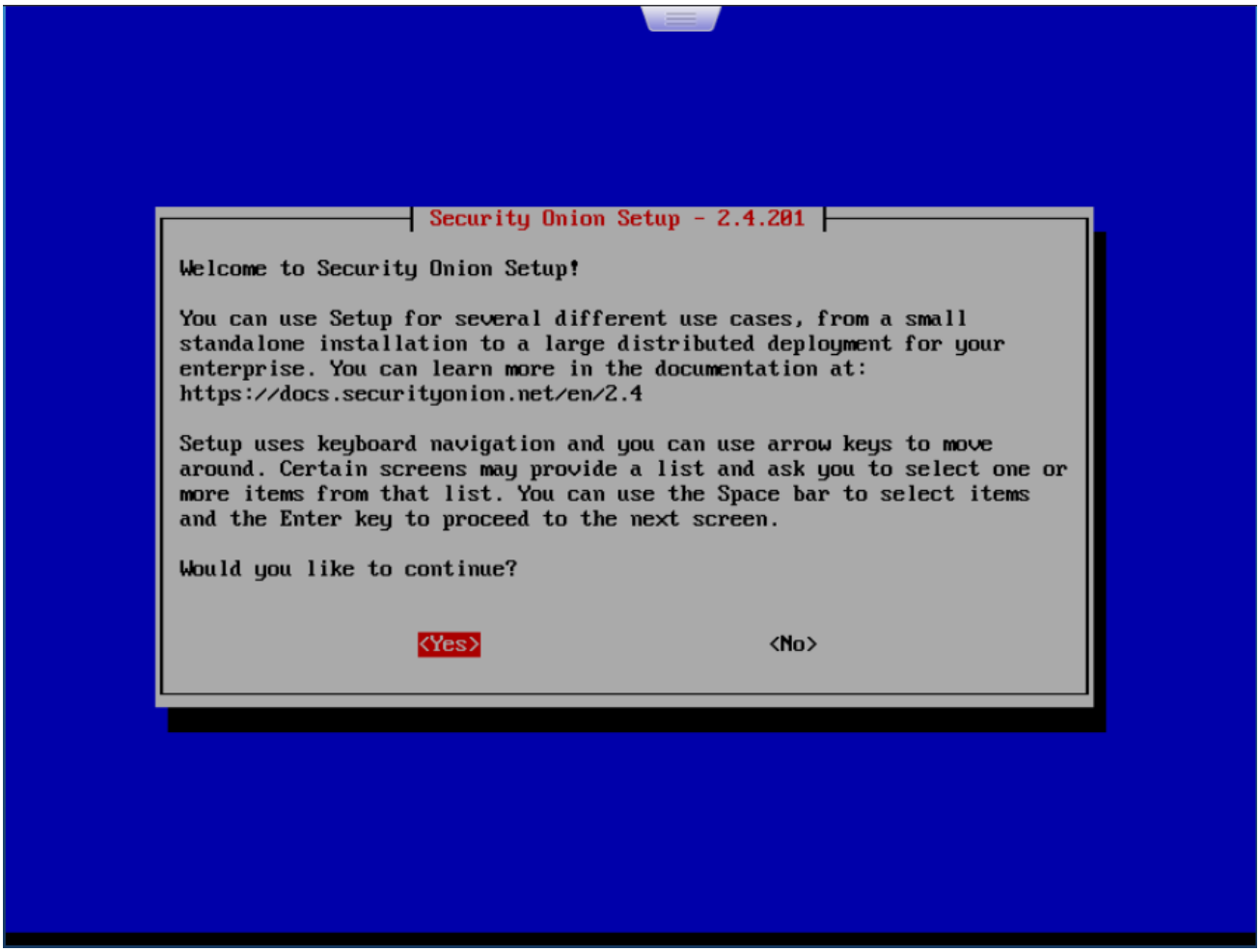
Once the admin username and password were entered, the installation was completed automatically. The installation configured the boot loader, network settings, and any required services. It then proceeded with the automatic reboot. Once the system was restarted, the ISO file was removed to enable the system to boot from the installed Security Onion environment.



STEP 5:

Starting Security Onion Setup

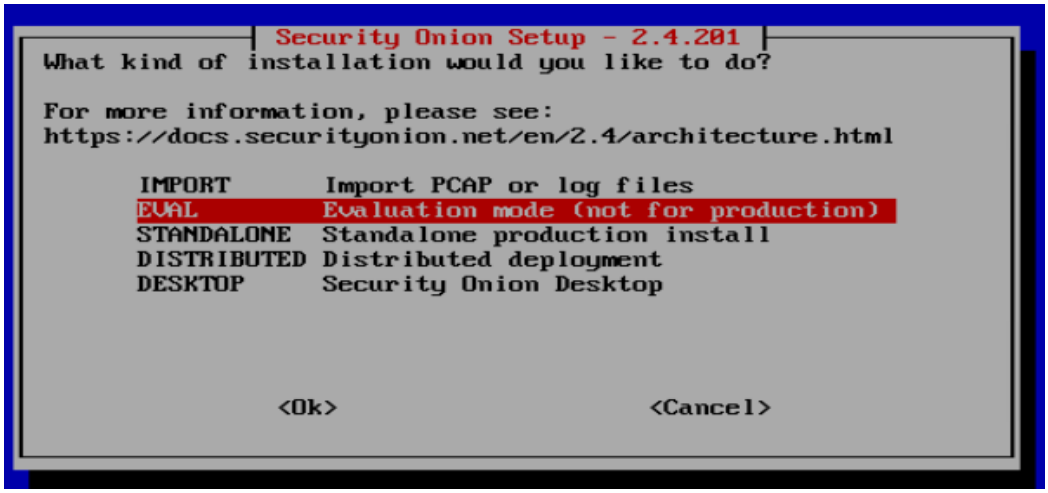
- After system reboot, logged in using the created administrative credentials.
- Executed the command sudo so-setup to start the configuration wizard.
- The Security Onion welcome screen appeared with setup information.
- Reviewed the deployment options (Standalone and Distributed).
- Selected “Yes” to proceed with the setup process.

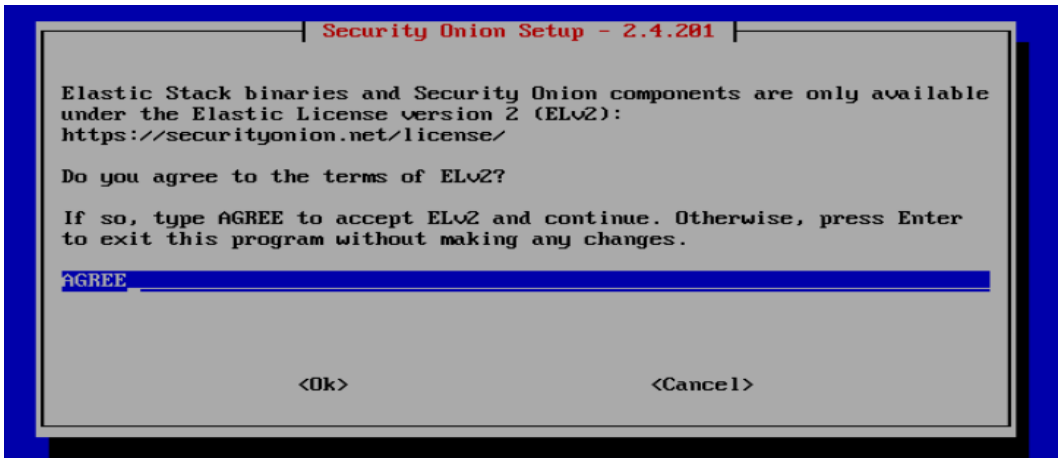


STEP 6 :

Selecting Installation Type

- Selected Install to run the standard Security Onion installation.
- Chose Evaluation Mode (EVAL) for lab and testing purposes.
- Evaluation mode is recommended for non-production environments.
- Read the Elastic License Version 2 (ELv2) agreement.
- Typed AGREE to accept the license terms and continue the setup.

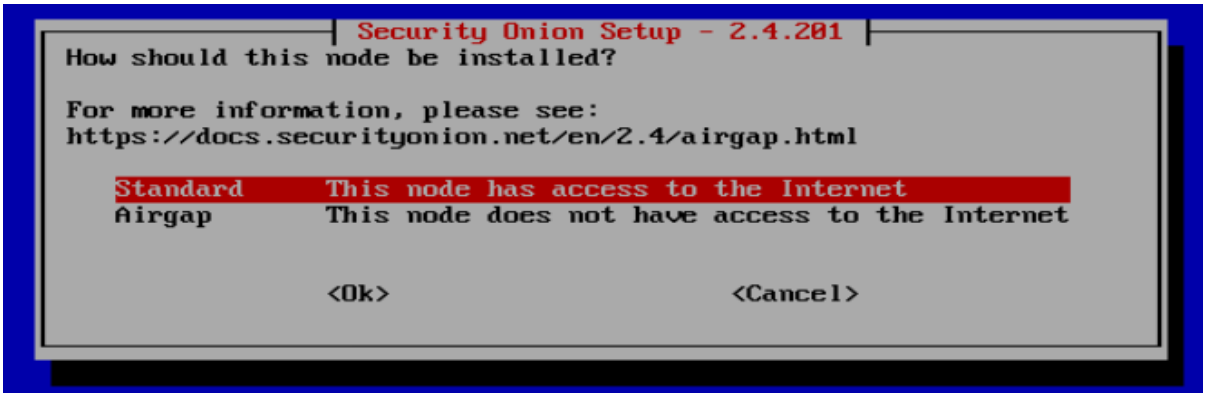


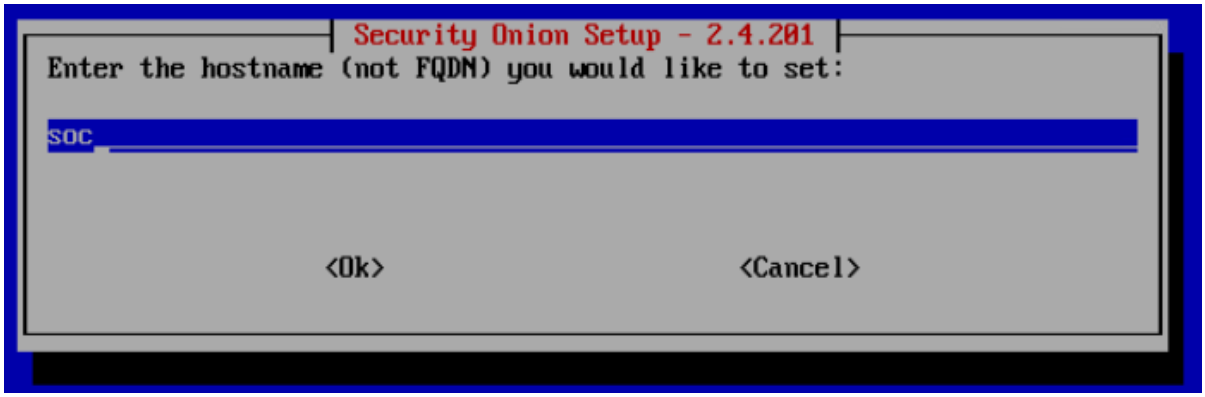


STEP 7:

Configuring Node Type, Hostname, and Network Interface

- Selected Standard installation (node has Internet access).
- Chose Standard instead of Airgap since the lab system has internet connectivity.
- Entered the hostname as SOC for system identification.
- Selected the primary network interface (ens192) for management.
- Confirmed the selected NIC (Link UP) to continue configuration.





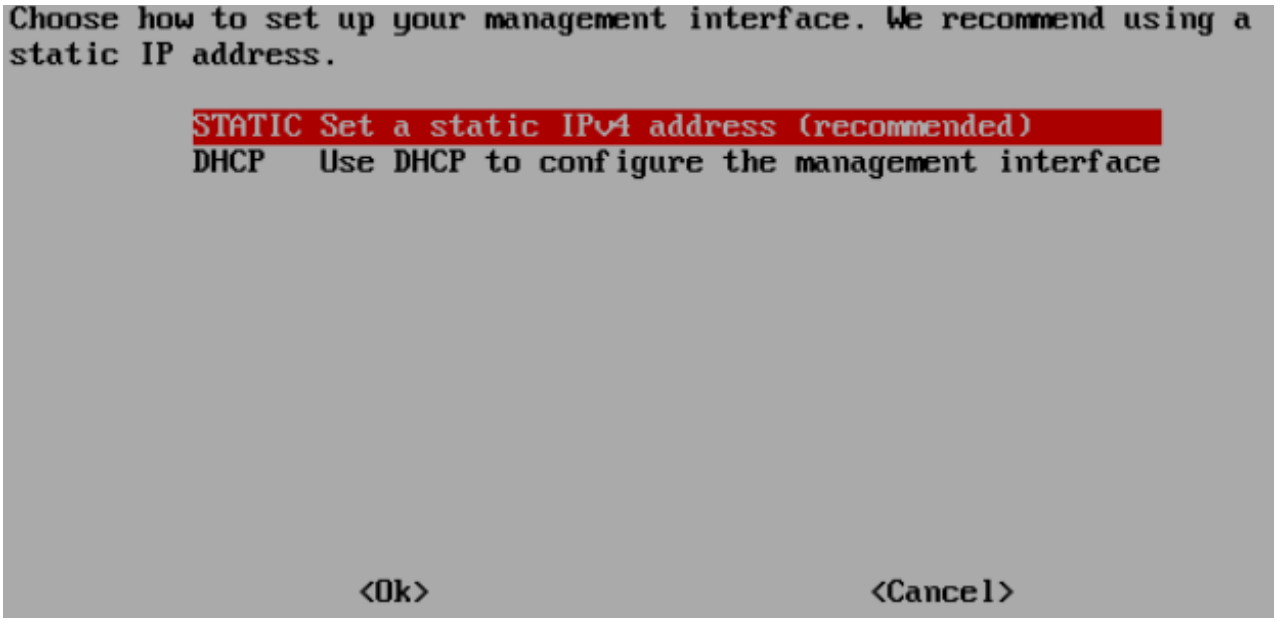
STEP 8:

Configuring Static IP Address and Network Settings

In this step, the management network interface of Security Onion was configured using a static IPv4 address. A static IP address is recommended because it ensures consistent and reliable access to the Security Onion management interface and web dashboard.

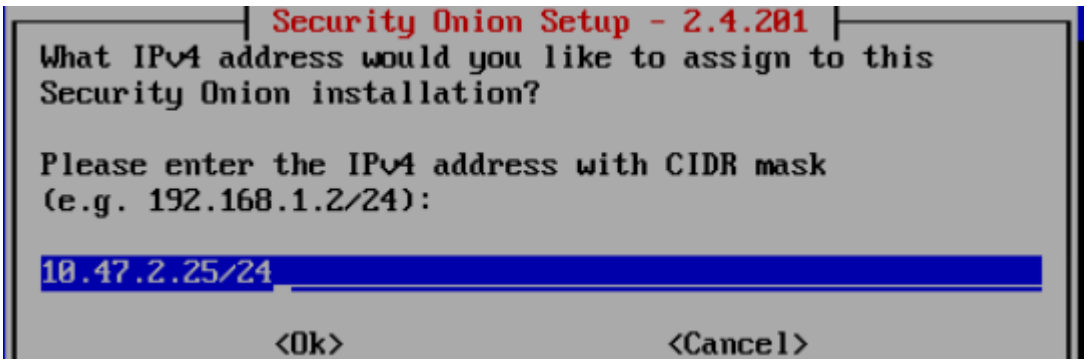
1. Selecting IP Configuration Method

- The setup wizard prompted to choose how the management interface should be configured.
- The "STATIC – Set a static IPv4 address (recommended)" option was selected.
- Static configuration is preferred in SOC environments to avoid IP address changes caused by DHCP.



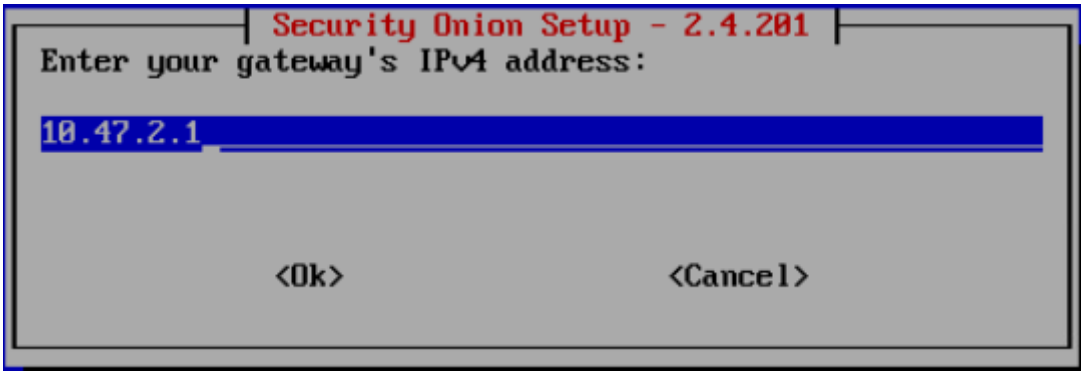
2. Assigning IPv4 Address

- The system requested an IPv4 address in CIDR format.
- The following IP address was assigned:
- 10.47.2.25 is the host IP address.



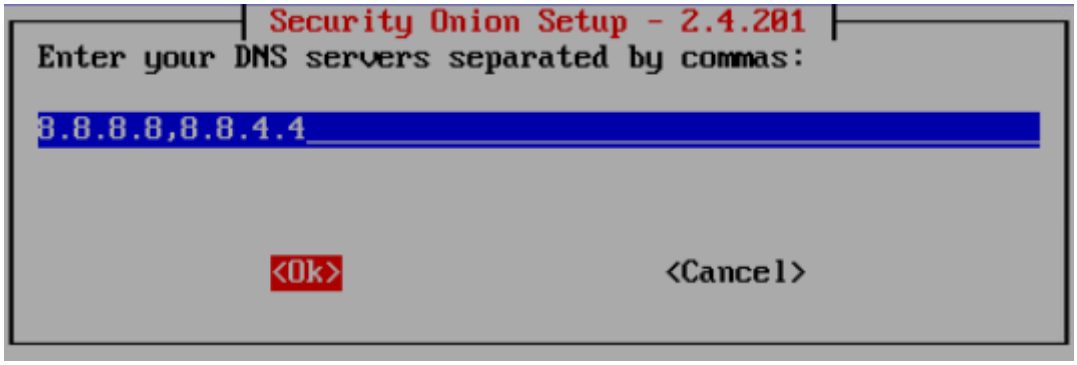
3. Configuring Default Gateway

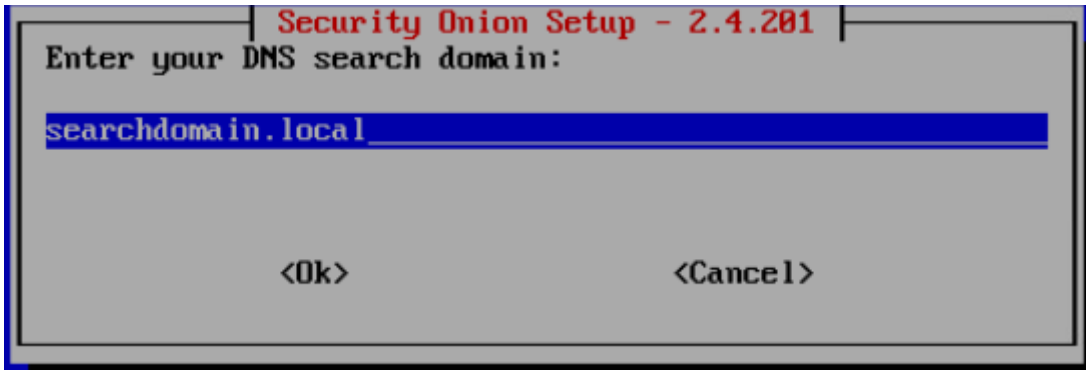
- The gateway IP address was entered as:



4. Configuring DNS Servers

- DNS server addresses were entered to enable domain name resolution.
- The following public DNS servers were configured:

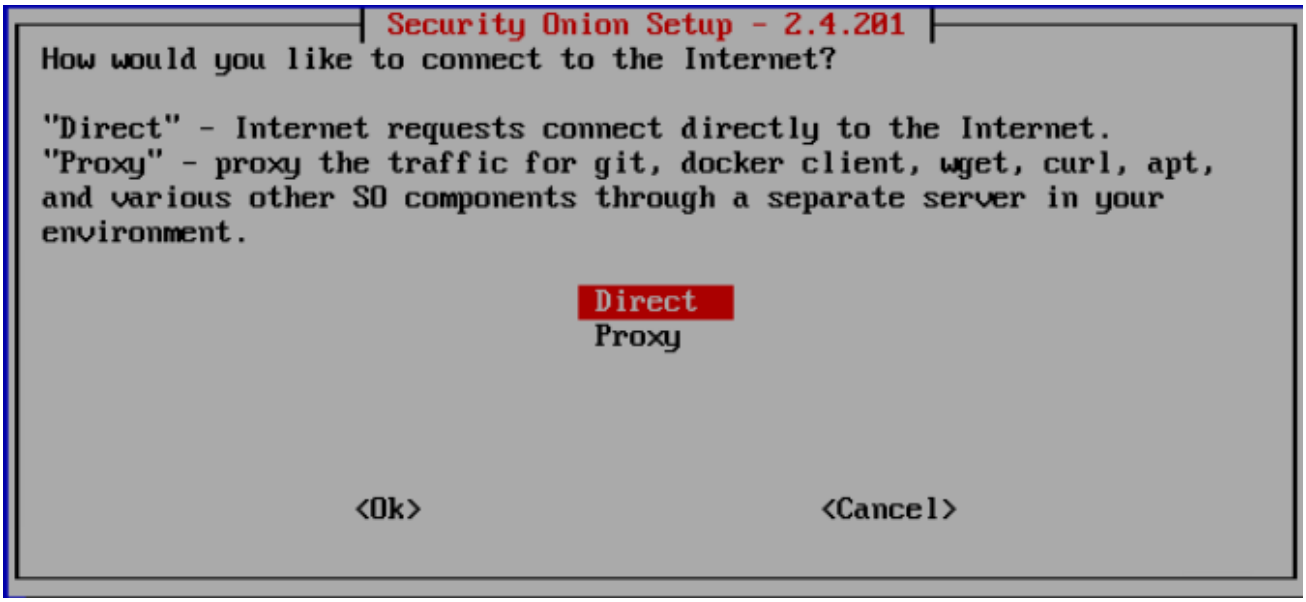


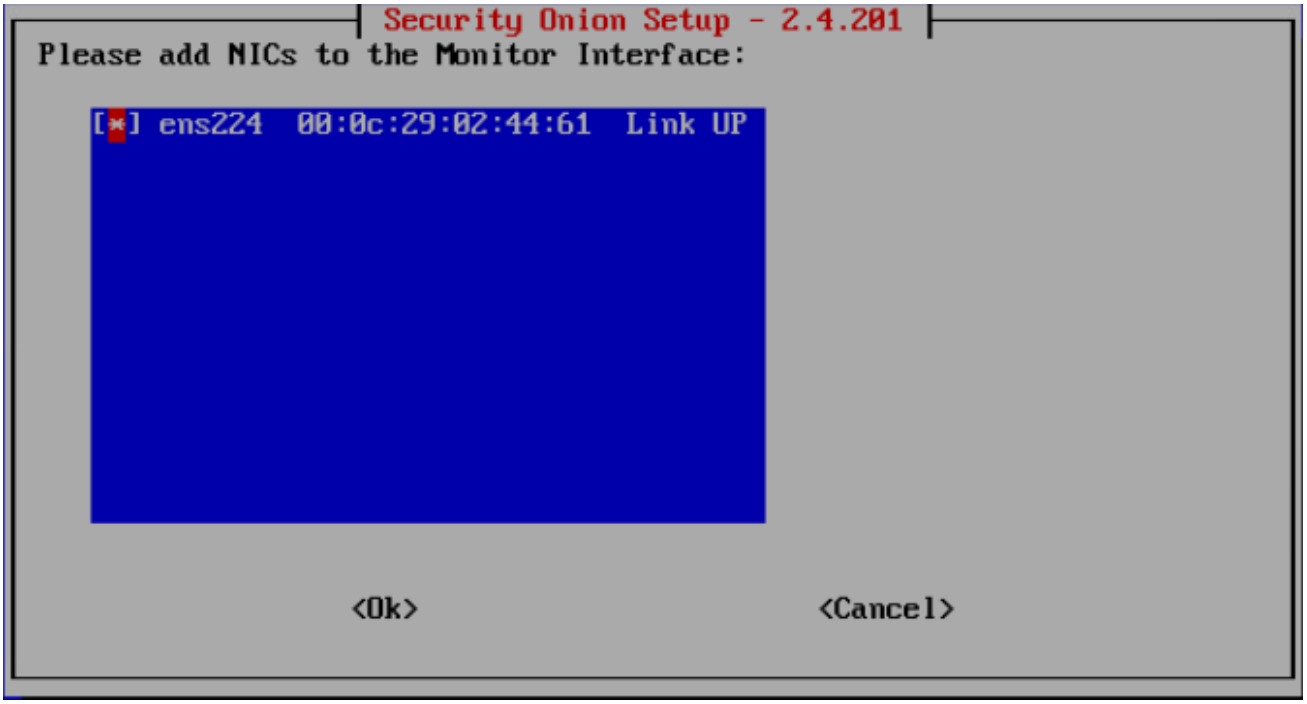


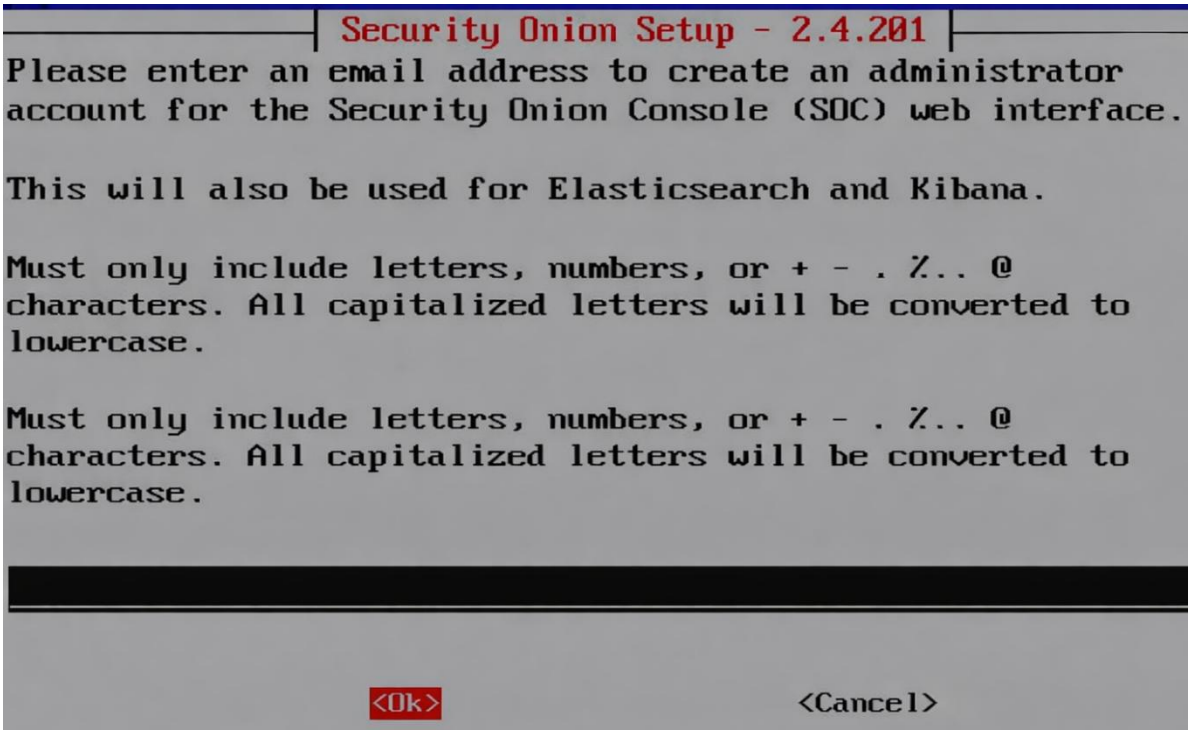
STEP 9:

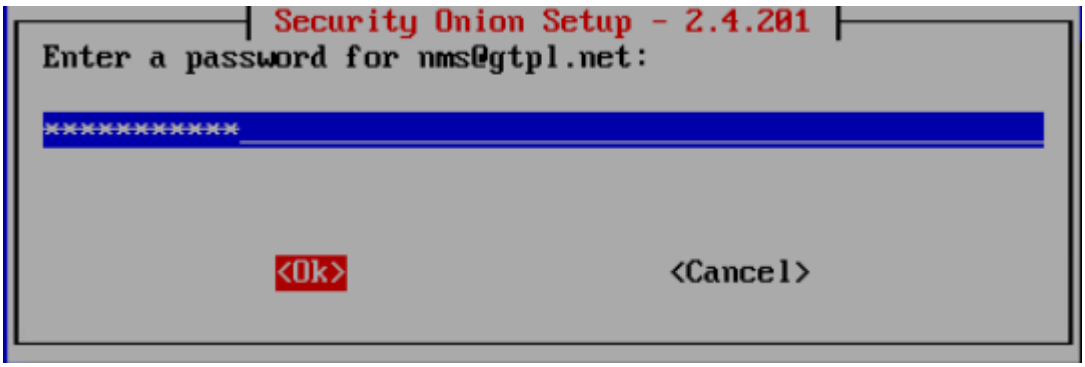
Internet, Monitor Interface & SOC Account Setup

- Selected Direct connection for internet access.
- Chose Direct because the lab environment has normal internet connectivity.
- Clicked OK to proceed.
- Selected ens224 as the Monitor Interface (sensor NIC).
- Enabled the interface to capture network traffic.
- Confirmed the NIC selection and continued.
- Entered an email address to create the SOC administrator account.
- This email is used for SOC web interface, Elasticsearch, and Kibana login.
- Set a strong password for the SOC admin account.
- Confirmed the password and selected OK to continue.

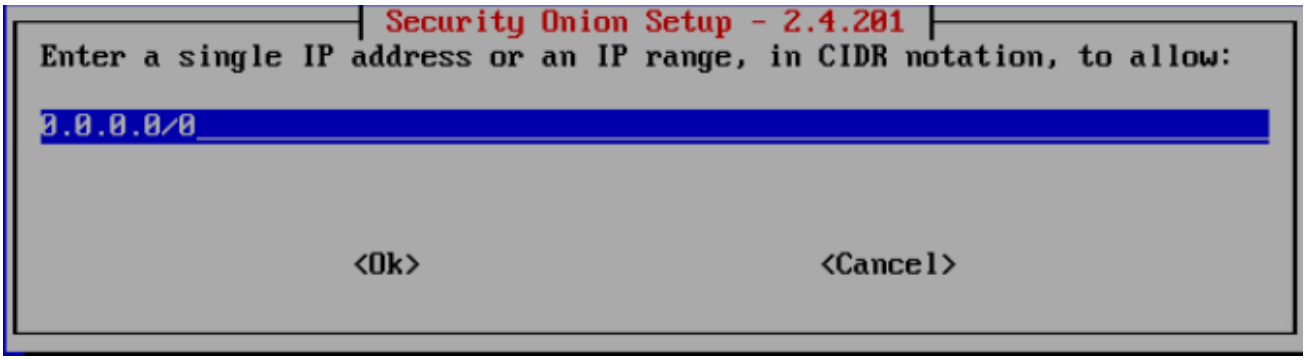


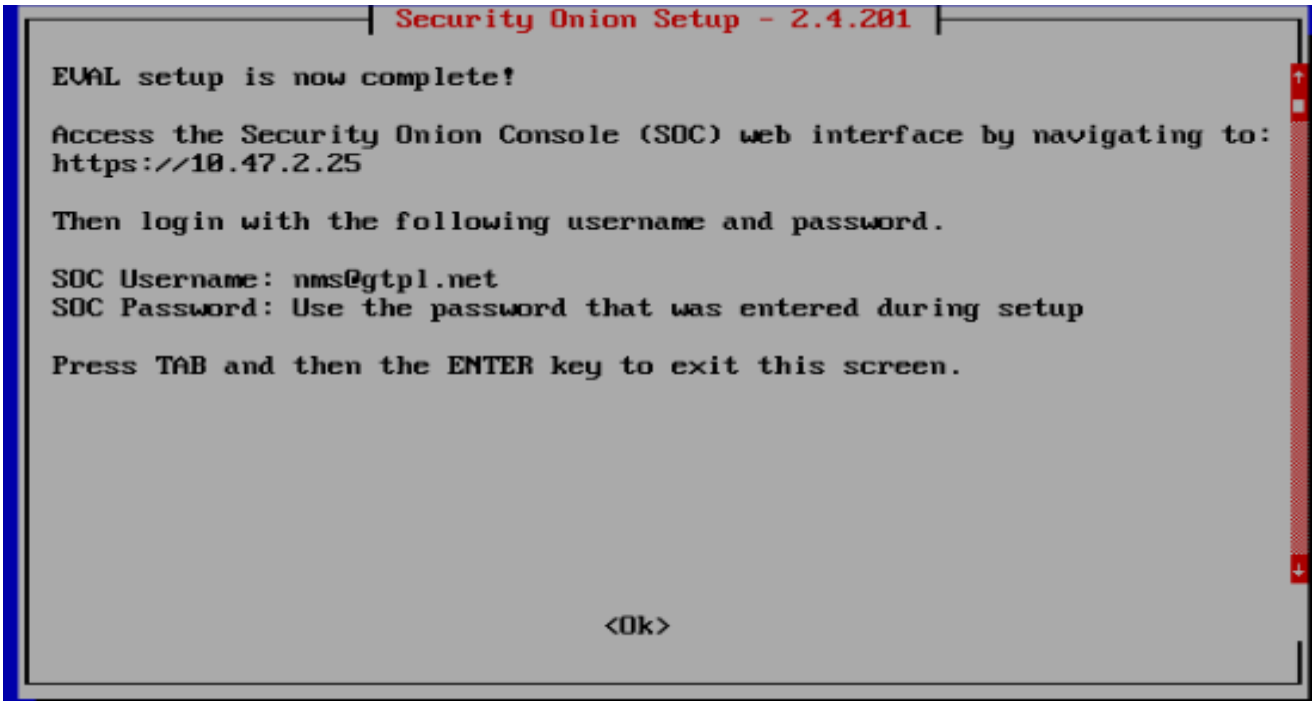

Security Onion Setup - 2.4.201
Please add NICs to the Monitor Interface:
[*] ens224 00:0c:29:02:44:61 Link UP
<Ok>
<Cancel>


Security Onion Setup - 2.4.201
Please enter an email address to create an administrator account for the Security Onion Console (SOC) web interface.
This will also be used for Elasticsearch and Kibana.
Must only include letters, numbers, or + - . %.. @ characters. All capitalized letters will be converted to lowercase.
Must only include letters, numbers, or + - . %.. @ characters. All capitalized letters will be converted to lowercase.
<Ok>
<Cancel>


Security Onion Setup - 2.4.201
Enter a password for nms@gtpl.net:
****
<Ok>
<Cancel>

STEP 10: Enter The IP Range To Allow and Give okay and exit





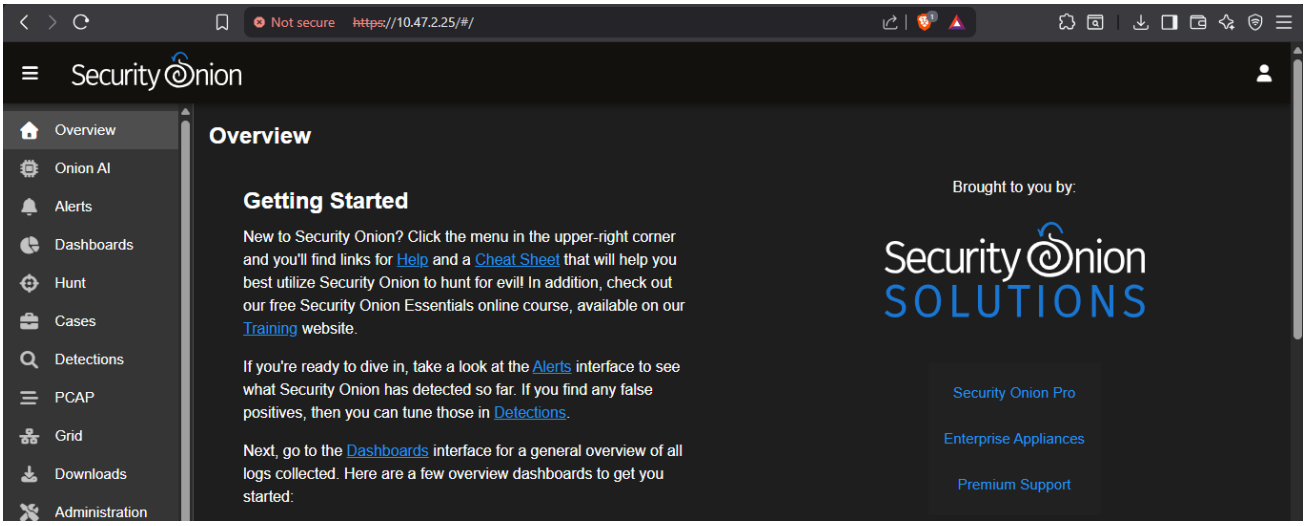
Enter the command in cli and check status of all services

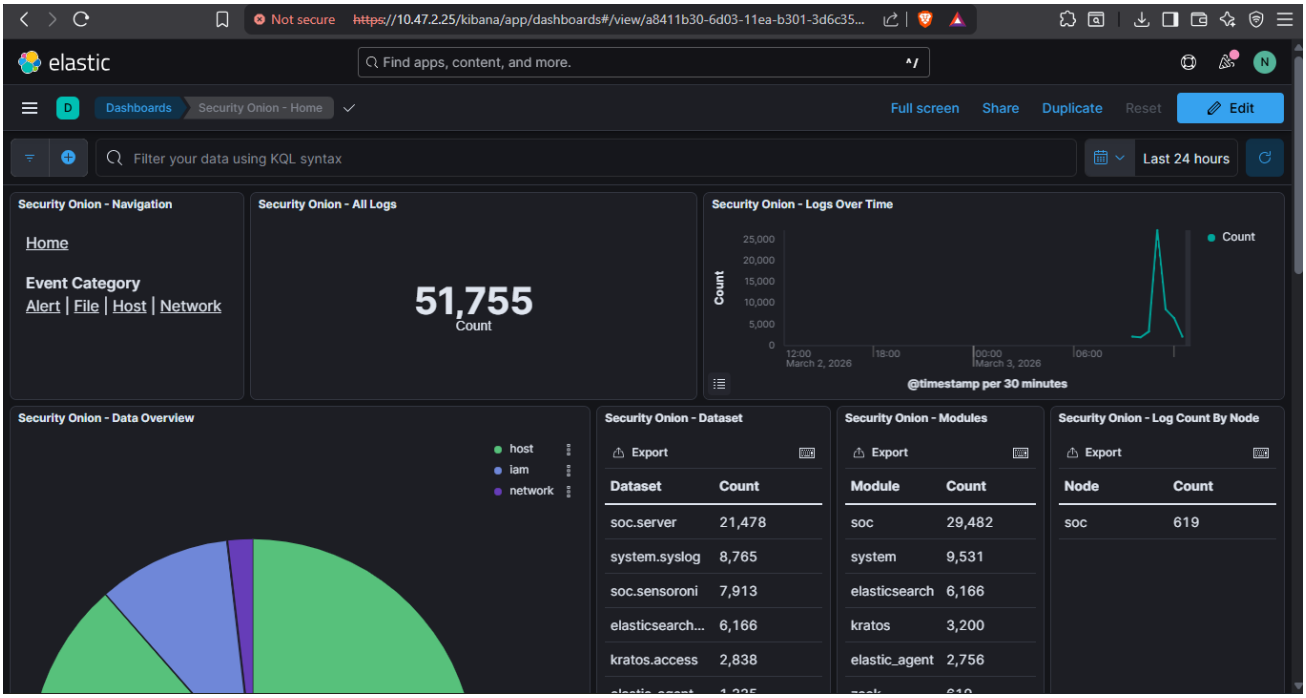
Command: sudo so-status

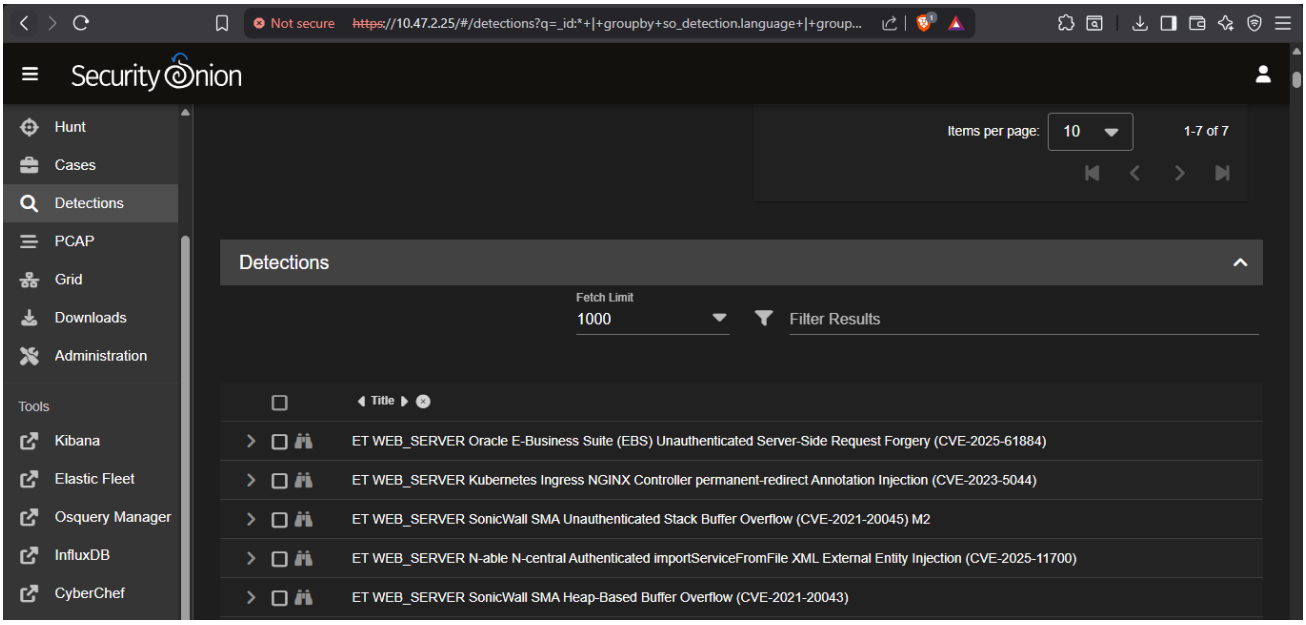
```
[soc-admin@localhost ~]$ sudo so-status
[sudo] password for soc-admin:

                    Security Onion Status
                 Container  | Status  |              Details
------------------------------------------------------------------
            so-dockerregistry | running |            Up 2 hours
                so-elastalert | running |            Up 2 hours
             so-elastic-fleet | running |            Up 2 hours
so-elastic-fleet-package-registry | running | Up 2 hours (healthy)
            so-elasticsearch | running |            Up 2 hours
                 so-influxdb | running | Up 2 hours (healthy)
                   so-kibana | running |            Up 2 hours
                   so-kratos | running |            Up 2 hours
                    so-nginx | running | Up 2 hours (healthy)
                so-sensoroni | running |            Up 2 hours
                      so-soc | running |            Up 2 hours
           so-strelka-backend | running |         Up 30 minutes
        so-strelka-coordinator | running |            Up 2 hours
         so-strelka-filestream | running |            Up 2 hours
           so-strelka-frontend | running |            Up 2 hours
         so-strelka-gatekeeper | running |            Up 2 hours
           so-strelka-manager | running |            Up 2 hours
                  so-suricata | running |         Up 31 minutes
                   so-telegraf | running |            Up 2 hours
                       so-zeek | running | Up 2 hours (healthy)

 ▪ This onion is ready to make your adversaries cry!

[soc-admin@localhost ~]$ _
```

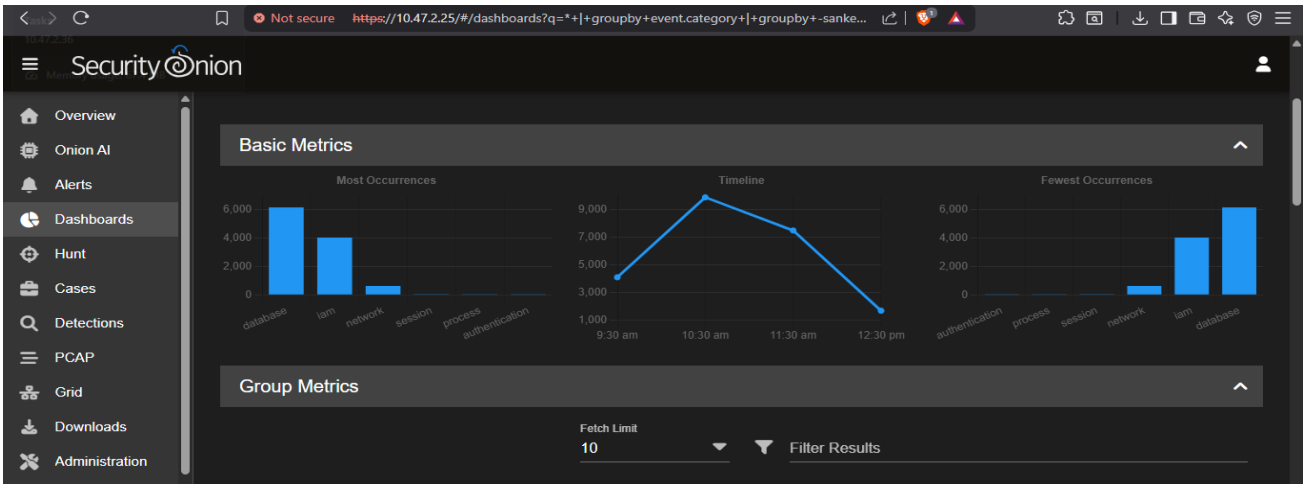
STEP11:

Enter the url in web browser **https://<host ip>** and check all tools are installed in security onion



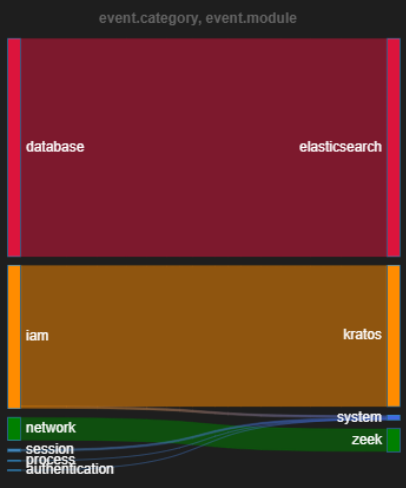

Not secure https://10.47.2.25/kibana/app/dashboards#/view/a8411b30-6d03-11ea-b301-3d6c35...
elastic
Find apps, content, and more.
Dashboards
Security Onion - Home
Full screen
Share
Duplicate
Reset
Edit
Filter your data using KQL syntax
Last 24 hours
Security Onion - Navigation
Home
Event Category
Alert | File | Host | Network
Security Onion - All Logs
51,755
Count
Security Onion - Logs Over Time
Count
@timestamp per 30 minutes
Security Onion - Data Overview
host
iam
network
Security Onion - Dataset
Export
Dataset Count
soc.server 21,478
system.syslog 8,765
soc.sensoroni 7,913
elasticsearch... 6,166
kratos.access 2,838
Security Onion - Modules
Export
Module Count
soc 29,482
system 9,531
elasticsearch 6,166
kratos 3,200
elastic_agent 2,756
Security Onion - Log Count By Node
Export
Node Count
soc 619


Not secure https://10.47.2.25/#/detections?q=_id:*+|+groupby+so_detection.language+|+group...
Security Onion
Hunt
Cases
Detections
PCAP
Grid
Downloads
Administration
Tools
Kibana
Elastic Fleet
Osquery Manager
InfluxDB
CyberChef
Items per page: 10
1-7 of 7
Detections
Fetch Limit
1000
Filter Results
Title
ET WEB_SERVER Oracle E-Business Suite (EBS) Unauthenticated Server-Side Request Forgery (CVE-2025-61884)
ET WEB_SERVER Kubernetes Ingress NGINX Controller permanent-redirect Annotation Injection (CVE-2023-5044)
ET WEB_SERVER SonicWall SMA Unauthenticated Stack Buffer Overflow (CVE-2021-20045) M2
ET WEB_SERVER N-able N-central Authenticated importServiceFromFile XML External Entity Injection (CVE-2025-11700)
ET WEB_SERVER SonicWall SMA Heap-Based Buffer Overflow (CVE-2021-20043)


Not secure https://10.47.2.25/#/dashboards?q=*+|+groupby+event.category+|+groupby+-sanke...
Security Onion
Overview
Onion AI
Alerts
Dashboards
Hunt
Cases
Detections
PCAP
Grid
Downloads
Administration
Basic Metrics
Most Occurrences
Timeline
Fewest Occurrences
database
iam
network
session
process
authentication
9:30 am
10:30 am
11:30 am
12:30 pm
Group Metrics
Fetch Limit
10
Filter Results

Count	event.category
6,166	database
4,040	iam
632	network
51	session
3	process
2	authentication

Items per page: 10 1-6 of 6



Count	event.module
9,559	system
6,166	elasticsearch
3,984	kratos
2,793	elastic_agent
632	zeek
1	sigma

Items per page: 10 1-6 of 6

Here is the step-by-step workflow to simulate and detect the attack.:

Phase 1: The Attack (Kali Linux or Attacker VM)

You need to generate enough "noise" for the Intrusion Detection System (IDS) to pick it up. Standard pings might be ignored, so we will use more aggressive Nmap flags.

1. **Identify the Target:** Ensure your Security Onion "Management" or "Sniffing" interface IP is reachable.
2. **Run an Aggressive Scan:** Run the following command from your attacker machine:

COMMAND: sudo nmap -A -T4 -p- [Target_IP]

- -A: Enables OS detection, version detection, script scanning, and traceroute (very noisy).
- -T4: Speeds up the execution (more likely to trigger threshold alerts).
- -p-: Scans all 65,535 ports.

```
┌──(root㉿kali)-[/home/kali]
└─# sudo nmap -A -T4 -p- 10.47.2.25
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-05 14:40 IST
Stats: 0:00:45 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 25.31% done; ETC: 14:42 (0:01:37 remaining)
Stats: 0:00:46 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 26.30% done; ETC: 14:42 (0:01:35 remaining)
Nmap scan report for 10.47.2.25
Host is up (0.00054s latency).
Not shown: 65532 filtered tcp ports (no-response)
PORT    STATE SERVICE  VERSION
22/tcp  open  ssh      OpenSSH 8.7 (protocol 2.0)
| ssh-hostkey:
|   256 81:6e:20:19:89:d7:49:ba:15:4b:20:70:6a:f8:d0:b2 (ECDSA)
|_  256 03:a0:84:1f:62:b8:b2:8e:41:6d:c8:ad:a2:c8:c5:ba (ED25519)
80/tcp  open  http     nginx
|_http-title: Did not follow redirect to https://10.47.2.25/
443/tcp open  ssl/http nginx
| ssl-cert: Subject: commonName=soc/stateOrProvinceName=Utah/countryName=US
| Subject Alternative Name: DNS:soc, IP Address:10.47.2.25, DNS:10.47.2.25
```

Phase 2: The Detection (Security Onion / Elasticsearch)

Once the scan finishes, Security Onion's **Suricata** (the IDS engine) should have flagged the traffic.

1. Check the Alerts Dashboard

- Log into your Security Onion Console (SOC).
- Click on **Alerts** in the left-hand sidebar.
- Look for signatures like:

Simulation 2:

Simulating a **Hydra** brute-force attack is the perfect follow-up to your Nmap scan. While Nmap finds the "doors" (ports), Hydra tries to "kick them down" by guessing passwords.

Since you are documenting this for your **GitHub project**, this simulation demonstrates how **Suricata** (alerts) and **Zeek** (connection logs) work together in Security Onion.

Phase 1: The Attack (Attacker VM)

Hydra is very noisy. We will target the **SSH** service (Port 22), as it's the most common target for brute forcing.

1. **Create a small password list:** (On your Kali/Attacker machine)
2. **Run the Hydra Attack:** Run this command against your target (the machine Security Onion is monitoring):

   hydra -l root -P passlist.txt -t 4 -V ssh://[TARGET_IP]

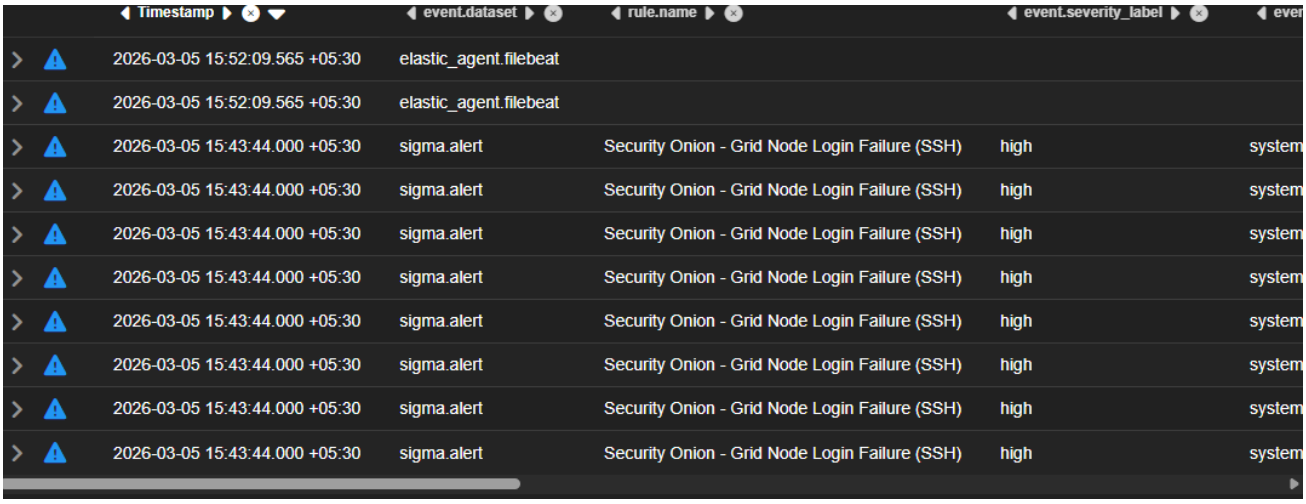
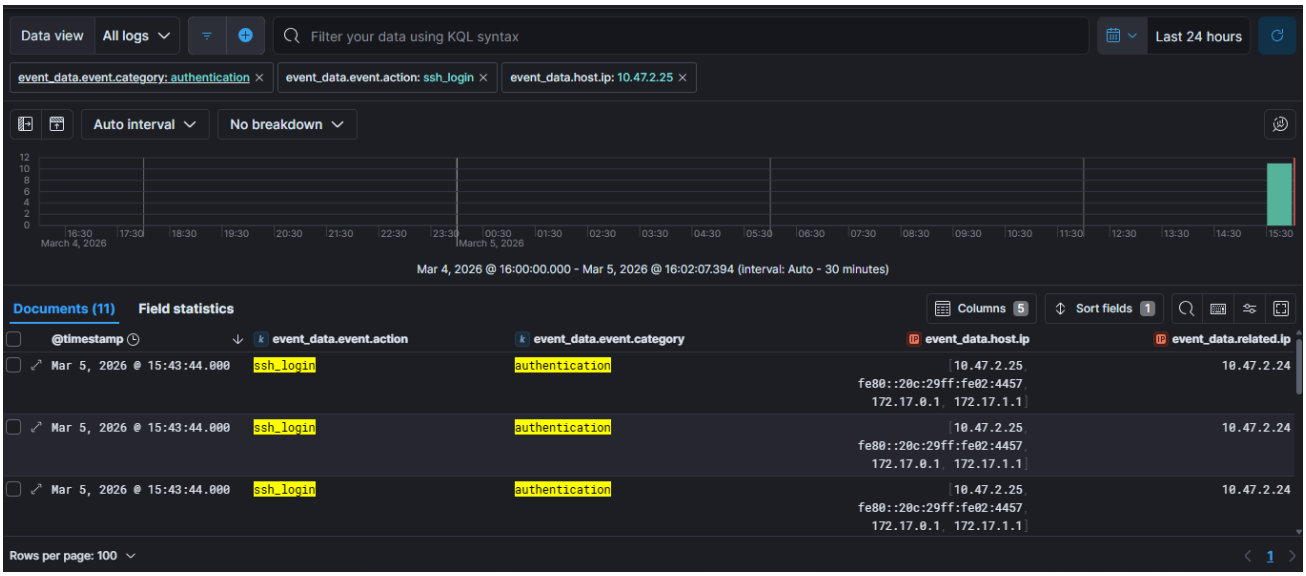
- -l root: The username you are targeting.
- -P passlist.txt: The list of passwords to try.
- -t 4: Uses 4 parallel threads (this speed usually triggers IDS alerts).
- -V: Verbose mode (shows you each attempt).

Phase 2: The Detection (Security Onion Console)

1. Identify Suricata Alerts

Go to the **Alerts** tab in your Security Onion Console. You should see high-severity alerts such as:

- ET SCAN Potential SSH Scan
- GPL SCAN SSH brute force
- ET BRUTEFORCE SSH login attempt



Timestamp	event.dataset	rule.name	event.severity_label	ever
2026-03-05 15:52:09.565 +05:30	elastic_agent.filebeat			
2026-03-05 15:52:09.565 +05:30	elastic_agent.filebeat			
2026-03-05 15:43:44.000 +05:30	sigma.alert	Security Onion - Grid Node Login Failure (SSH)	high	system
2026-03-05 15:43:44.000 +05:30	sigma.alert	Security Onion - Grid Node Login Failure (SSH)	high	system
2026-03-05 15:43:44.000 +05:30	sigma.alert	Security Onion - Grid Node Login Failure (SSH)	high	system
2026-03-05 15:43:44.000 +05:30	sigma.alert	Security Onion - Grid Node Login Failure (SSH)	high	system
2026-03-05 15:43:44.000 +05:30	sigma.alert	Security Onion - Grid Node Login Failure (SSH)	high	system
2026-03-05 15:43:44.000 +05:30	sigma.alert	Security Onion - Grid Node Login Failure (SSH)	high	system
2026-03-05 15:43:44.000 +05:30	sigma.alert	Security Onion - Grid Node Login Failure (SSH)	high	system
2026-03-05 15:43:44.000 +05:30	sigma.alert	Security Onion - Grid Node Login Failure (SSH)	high	system